Orthogonal Complements and Orthonormal Matrices

Orthogonal Complements

Defn. *For a set W, the* ***orthogonal complement*** *denoted $W^{\perp}$ is the set of all vectors that are orthogonal to* ***all*** *of W.*

Orthogonal Complements are Subspaces

Fact. *For any subset W, the orthogonal complement $W^{\perp}$ is a subspace.*

This can be shown using the standard recipe for verifying a subspace.

A 3-D Example: Planes and Lines

Consider in $\mathbb{R}^3$ plane P given by $3x + 4y - z = 0$.

If we take any vector (x, y, z) in the plane P, it is orthogonal to vector $(3, 4, -1)$ (just compute their dot product!).

Thus orthogonal complement of plane P is line parallel to vector $(3, 4, -1)$.

The Row and Null Spaces are Orthogonal

Fact. *For any matrix A:*

$$(Row\ A)^{\perp} = Nul\ A \quad \textit{and} \quad (Col\ A)^{\perp} = Nul\ A^T$$

Testing in Orthogonal Complement

To check that vector $\boldsymbol{v}$ is orthogonal to all of W, it is sufficient that $\boldsymbol{v}$ is orthogonal to a basis of W.

Orthogonal and Orthonormal Sets

Defn. *An **orthogonal set** is a collection of vectors that are pairwise orthogonal. An **orthonormal set** is an orthogonal set of **unit** vectors.*

(The "***pairwise***" means that for every pair of vectors, the two vectors are orthogonal to each other.)

Orthogonal Implies Independence

Fact. *If S is an orthogonal set of nonzero vectors, then S is linearly independent.*

Proof idea: consider a linear combination of S that sums to zero and take its dot product with each element of S. When the dust settles, we see that each weight must be zero.

Orthonormal Matrices

Defn. *An **orthonormal matrix** has orthonormal columns and rows.*

Note $U^T U = I$ for orthonormal matrix U.

As a matrix transform, such a matrix preserves lengths and orthogonality.

Orthonormal Coordinate Systems

Fact. *If* $B = \{\boldsymbol{w}_i\}$ *is an orthonormal basis, then the coordinates of vector* $\boldsymbol{v}$ *in terms of* B *are the dot-products of* $\boldsymbol{v}$ *with each* $\boldsymbol{w}_i$*.*

A Space has an Orthonormal Basis

Fact. *Every vector space has an orthonormal basis.*

Summary

For a set W, the orthogonal complement denoted $W^\perp$ is the subspace of all vectors orthogonal to all of W. To check that vector in $W^\perp$, it suffices to check for a basis of W.

The row and null spaces are orthogonal complements. In $\mathbb{R}^3$ vector (a, b, c) is basis for space orthogonal to plane $ax + by + cz = 0$.

Summary (cont)

An orthogonal set is a collection of vectors that are pairwise orthogonal; an orthonormal set is an orthogonal set of unit vectors. An orthogonal set of nonzero vectors is linearly independent. Every vector space has an orthonormal basis.

An orthonormal matrix U has orthonormal columns and rows; equivalently, $U^T U = I$.